Random variable and its probability distributions

Random variable:

A variable whose values are any definite numbers or quantities that arise as a result of chance factors such that they can not exactly be predicted in advance, is called random variable.

- A random variable is a way to quantify the results of a random event.
- It maps the outcomes of a random experiment to numerical values.
- For example, when flipping a coin, the outcome (heads or tails) can be assigned the numerical values 1 and 0, respectively, making it a random variable.

A random variable is a real-valued function defined over a sample space.

Discrete random variable:

A random variable define over a discrete sample space (that may only take on a finite or countable number of different isolated values) is referred to as a discrete random variable.

Some examples of discrete random variables:

Number of heads in coin tosses.

Number of defective items in a batch.

Number of cars crossing a bridge in an hour.

Number of customers in a store.

Number of sixes in multiple dice rolls.

Number of lottery tickets purchased.

Number of students in a class.

Number of books on a shelf.

Continuous random variable:

A random variable define over a continuous sample space (which may take any value in a certain interval or collection of intervals) is referred to as a discrete random variable.

Examples:

The height of a student in a class.

The temperature of a room.

The amount of time it takes to complete a task.

The weight of a randomly selected apple.

Distinction from Discrete Random Variables:

Discrete Random Variables: Can only take on specific, separate values (like the number of students in a class or the results of a coin flip).

Continuous Random Variables: Can take on any value within a range.

Probability distribution:

Any statement of a function associating each of a set of mutually exclusive and exhaustive classes or class intervals with its probability is a probability distribution.

Related terms:

Random Variable: A variable whose value is a numerical outcome of a random phenomenon.

Sample Space: The set of all possible outcomes of an experiment.

Probability Mass Function (PMF): For discrete random variables, the PMF gives the probability of each possible value.

Probability Density Function (PDF): For continuous random variables, the PDF describes the probability density at each point, and the area under the curve over a range represents the probability of the variable falling within that range.

Cumulative Distribution Function (CDF): Gives the probability that a random variable is less than or equal to a certain value.

Probability distributions are used in various fields:

Statistics: Describing data, hypothesis testing, and making predictions.

Finance: Modeling stock prices, assessing investment risk.

Science: Analyzing experimental data, predicting outcomes in simulations.

Engineering: Reliability analysis, quality control.

Actuarial Science: Pricing insurance policies, assessing risk.

Discrete probability distribution:

If a random variable X has a discrete distribution, the probability distribution of X is defined as the function f such that for any real number x, $f(x) = P(X = x)$.

It is important to emphasize that all functions are not probability functions. The function $f(x)$ defined above must satisfy the following conditions in order to be a probability function:

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $P(X = x) = f(x)$

Example-1:

Verify if the following functions are probability functions.

(a) $f(x) = \frac{2x-1}{8}$, $\quad x = 0, 1, 2, 3$

(b) $f(x) = \frac{x+1}{16}$, $\quad x = 0, 1, 2, 3$

Solution:

Summing the function over the entire range of X,

(a) $\sum_{x=0}^{3} f(x) = f(0) + f(1) + f(2) + f(3)$
$= -\frac{1}{8} + \frac{1}{8} + \frac{3}{8} + \frac{5}{8}$

$= 1$

But $P(X = 0) = f(0) = -\frac{1}{8}$, which contradicts the condition (1).

(a) $\sum_{x=0}^{3} f(x) = f(0) + f(1) + f(2) + f(3)$
$= \frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16}$

$= \frac{10}{16} \neq 1$

Hence it is not a probability function although for all values of x, $f(x) \geq 0$.

Example-2:

A bag contains 10 balls of which 4 are black. If 3 balls are selected at random without replacement, obtain the probability distribution for the number of black balls drawn.

Solution:

If X denotes the number of black balls drawn, then clearly X can assume values $0, 1, 2$ and 3. To obtain the probability distribution of X, we need to compute the probabilities associated with $0, 1, 2$ and 3. Since 3 balls are to be chosen, the number of ways in which this choice can be made is $C_{(10,3)}$.

Thus

$f(0) = P(X = 0) = \frac{C(4,0) \times C(6,3)}{C(10,3)} = \frac{20}{120}$.

$f(0) = P(X = 0) = \frac{C(4,1) \times C(6,2)}{C(10,3)} = \frac{60}{120}$.

$f(0) = P(X = 0) = \frac{C(4,2) \times C(6,1)}{C(10,3)} = \frac{36}{120}$.

$f(0) = P(X = 0) = \frac{C(4,3) \times C(6,0)}{C(10,3)} = \frac{4}{120}$.

Hence the tabular form of the probability distribution of Xwill be as follows:

x	0	1	2	3
$f(x)$	$^{20}/_{120}$	$^{60}/_{120}$	$^{36}/_{120}$	$^{4}/_{120}$

A close view of the calculations reveals that the probabilities corresponding to different values of X can be computed using the following formula:

$$f(x) = \frac{C(4,x)\times C(6,3-x)}{C(10,3)}, \qquad x = 0, 1, 2, 3.$$

Example-3:

The probability function of a discrete random variable X is

$$f(x) = \alpha(\frac{3}{4})^x, \qquad x = 0, 1, 2, 3, \dots \dots \dots \dots \dots \infty$$

$$= 0, \qquad \text{elsewhere}$$

Evaluate $\boldsymbol{\alpha}$ and find $P(X \leq 3)$.

Solution:

Since $f(x)$ is probability function, $\sum_0^\infty f(x) = 1$.

But $f(0) = \alpha(\frac{3}{4})^0 = \alpha,\ f(1) = \alpha(\frac{3}{4})^1 = \alpha(\frac{3}{4}),\ f(2) = \alpha(\frac{3}{4})^2$ and $f(3) = \alpha(\frac{3}{4})^3$ and so on.

Hence $\sum_0^\infty f(x) = \alpha + \alpha\left(\frac{3}{4}\right) + \alpha(\frac{3}{4})^2 + \alpha(\frac{3}{4})^3 + \cdots \dots \dots \dots \dots \dots \dots$

$$= \alpha(1 + \left(\frac{3}{4}\right) + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \cdots \dots \dots \dots \dots \dots \dots)$$

$$= \alpha(\frac{1}{1-\frac{3}{4}})$$

$$= 4\alpha$$

Thus $\alpha = \frac{1}{4}$.

Also $P(X \leq 3) = \frac{1}{4}(1 + \frac{3}{4} + \frac{9}{16} + \frac{127}{64})$

$$= \frac{175}{256}.$$

Discrete distribution function:

The cumulative distribution function (CDF) or simply the distribution function $F(x)$ of a discrete random variable X with probability function $f(x)$ define over all real numbers x is the cumulative probability upto and including the point x. Symbolically, it defined as follows:

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

Example-4:

Consider the probability distribution of the random variable X in **example-2**

x	0	1	2	3
$f(x)$	$^{20}/_{120}$	$^{60}/_{120}$	$^{36}/_{120}$	$^{4}/_{120}$

Here

$$F(0) = P(X = 0) = f(0) = \frac{20}{120}$$

$$F(1) = P(X \leq 1) = f(0) + f(1) = \frac{20}{120} + \frac{60}{120} = \frac{80}{120}$$

$$F(2) = P(X \leq 2) = f(0) + f(1) + f(2) = \frac{20}{120} + \frac{60}{120} + \frac{36}{120} = \frac{116}{120}$$

$$F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3) = \frac{20}{120} + \frac{60}{120} + \frac{36}{120} + \frac{4}{120} = 1$$

A tabular presentation of $F(x)$ is as follows:

x	< 0	*0*	*1*	*2*	*3*	> 3
f(x)	*0*	$\frac{20}{120}$	$\frac{60}{120}$	$\frac{36}{120}$	$\frac{4}{120}$	*0*
F(x)	*0*	$\frac{20}{120}$	$\frac{80}{120}$	$\frac{116}{120}$	*1*	*1*

A formal way of presenting the distribution function $F(x)$is as follows:

$$F(x) = \begin{cases} 0, & for\ x < 0 \\ \frac{20}{120}, & for\ 0 \leq x < 1 \\ \frac{80}{120}, & for\ 1 \leq x < 2 \\ \frac{116}{120}, & for\ 2 \leq x < 3 \\ 1, & for\ x \geq 3 \end{cases}$$

Continuous probability distribution:

A probability density function is a non-negative function and is constructed so that the area under its curve bounded by the x-axis is equal to unity when computed over the range of x, for which $f(x)$ is defined.

The above definition leads to conclude that a pdf is one that possesses the following properties:

1. $f(x) \geq 0$.
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_{a}^{b} f(x)dx$

Example-1:

A random variable *X* has the following functional form:

$$f(x) = \begin{cases} kx, & 0 < x < 4 \\ 0, & elsewhere \end{cases}$$

(i) Determine k for which $f(x)$ is a density function.

(ii) Find $P(1 < X < 2)$ and $P(X > 2)$

Solution:

(i) For *X* to be a density function, we must have $\int_{-\infty}^{\infty} f(x)dx = 1$

Thus

$$k\int_{0}^{4} xdx = 1$$

Or, $k(\frac{x^2}{2})_0^4 = 1$

Or, $8k = 1$,

From which $k = \frac{1}{8}$

(ii) $P(1 < X < 2) = \frac{1}{8}\int_{1}^{2} xdx$

$$= \frac{1}{8}(\frac{x^2}{2})_1^2$$

$$= \frac{3}{16}$$

And $P(X > 2) = \frac{1}{8}\int_{2}^{4} xdx$

$$= \frac{1}{8}(\frac{x^2}{2})_2^4$$

$$= \frac{3}{4}$$

Example-2:

A random variable *X* has the following density function:

$$f(x) = \begin{cases} \frac{2}{27}(1+x), & 2 < x < 5 \\ 0, & elsewhere \end{cases}$$

a) Verify that it satisfies the condition $\int_{-\infty}^{\infty} f(x)dx = 1$
b) Find $P(X < 4)$
c) Find $P(3 < X < 4)$

Continuous distribution function:

The cumulative distribution or distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is defined as

$$F(x) = \int_{-\infty}^{x} f(t)dt = 1$$

$$= P(X \leq x)$$

Example-3:

A random variable X has the following density function:

$$f(x) = \begin{cases} \frac{2}{27}(1+x), & 2 < x < 5 \\ 0, & elsewhere \end{cases}$$

Find $F(3)$ and $F(4)$ and hence verify that $P(3 < X < 4) = F(4) - F(3)$.

Solution:

$$F(3) = P(X < 3) = \frac{2}{27}\int_{2}^{3}(1+x)dx$$

$$= \frac{2}{27}(x + \frac{x^2}{2})_2^3$$

$$= \frac{7}{27}$$

$$F(4) = P(X < 4) = \frac{2}{27}\int_{2}^{4}(1+x)dx$$

$$= \frac{2}{27}(x + \frac{x^2}{2})_2^4$$

$$= \frac{16}{27}$$

Hence

$$F(4) - F(3) = \frac{16}{27} - \frac{7}{27}$$

$$= \frac{1}{3}$$

Now, $P(3 < X < 4) = \frac{2}{27}\int_3^4 (1+x)dx$

$$= \frac{2}{27}(x + \frac{x^2}{2})_3^4$$

$$= \frac{1}{3}$$

Hence

$P(3 < X < 4) = F(4) - F(3)$.

Example-4:

Find the constant k so that the following function may be a density function:

$$f(x) = \begin{cases} \frac{1}{k}, & a \le x \le b \\ 0, & elsewhere \end{cases}$$

Find also the cumulative distribution of the random variable *X* defined above.

Example-5:

Following are the two cumulative distribution function of *X*,

$$F(x) = \begin{cases} 0, & x < -2 \\ \frac{1}{12}(2+x), & -2 \le x < 0 \\ \frac{1}{6}(1+x), & 0 \le x < 4 \\ \frac{1}{12}(2+x), & 4 \le x < 6 \\ 1, & x \ge 6 \end{cases}$$

Find $f(x)$.